Quick Start Guide (Must Read)

1. Documentation Directory

Name	Date modified	Type
1. Tutorials	26/08/2025 17:52	File folder
2. Software	23/08/2025 15:15	File folder
3. Technical Parameter & Diagram	28/08/2025 18:20	File folder

1. **Tutorials:** Includes basic operation guides and advanced feature tutorials for the robot. Suitable for users at different levels to get started quickly. Provides step-by-step guides and sample projects to help beginners learn from scratch.
2. **Software:** Offers all the essential software for robot development, debugging, and control, including PC software, mobile app, firmware, and flashing tools. Contains the robot's core control programs, algorithm implementations, and sensor drivers with fully open and editable source code.
3. **Technical Parameter & Diagram:** Provides complete hardware design resources such as circuit schematics, 3D structural models, and engineering drawings for debugging.
4. **FAQ:** Compiles frequently asked questions, offering troubleshooting steps, error code explanations, optimization tips, and other solutions. Continuously updated to cover most usage scenarios.

2. Tutorials Directory (Key Sections)

The following is the course content directory included in 1. Tutorials.

Name	Date modified	Type
1. Read First	20/08/2025 15:20	File folder
2. App Control	20/08/2025 15:20	File folder
3. PC Control and Action Group Editing	26/08/2025 18:20	File folder
4. Arduino IDE Installation & Servo Calibr...	20/08/2025 15:21	File folder
5. Motion Control Basic Course	20/08/2025 15:20	File folder
6. Secondary Development Course	20/08/2025 15:36	File folder
7. AI Vision Feature Course	20/08/2025 16:02	File folder
8. AI Voice Interaction Course	20/08/2025 16:02	File folder
9. Serial Communication Hands-on Course	20/08/2025 16:02	File folder

1. **Read First:** Provides an introduction to the robot's basic functions, interface descriptions, user guide, and important safety notes.

Name	Date modified	Type
1.1 Introduction to miniHexa.docx	19/08/2025 09:39	DOCX 文档
1.2 Battery Charging and First Startup.docx	14/08/2025 14:47	DOCX 文档
1.3 miniHexa Interface Overview.docx	18/08/2025 11:03	DOCX 文档

2. **App Control:** Includes the dedicated mobile application (app) for controlling the robot and the corresponding user manual. Supports remote control and real-time monitoring of device status. Features include robot movement, camera feed transmission, and more. Source code for app control is provided with full modification permissions, allowing users to customize development.

Name	Date modified	Type
2.2 App Installation Package	28/08/2025 18:22	File folder
2.3 App Program Files	23/08/2025 14:25	File folder
2.1 App User Guide.docx	20/08/2025 14:47	DOCX 文档

3. **PC Control and Action Group Editing:** Provides a complete software suite required for robot development, including PC software, mobile applications, firmware, and flashing tools. Includes core control programs, algorithm implementation code, and sensor drivers, which are all open-source and modifiable. Covers usage of the PC software for editing motion groups, and you can create your own custom robot motion sequences.

Name	Date modified	Type
3.2 PC Software Package	26/08/2025 17:52	File folder
3.3 PC Software Program Files	26/08/2025 17:52	File folder
3.1 Introduction to PC Software.docx	26/08/2025 16:25	DOCX 文档
3.1 Introduction to PC Software.pdf	26/08/2025 17:56	WPS PDF 文档
3.4 Omnidirectional Movement Control....	26/08/2025 16:34	DOCX 文档
3.4 Omnidirectional Movement Control....	26/08/2025 17:55	WPS PDF 文档
3.5 Posture Control.docx	26/08/2025 16:42	DOCX 文档
3.5 Posture Control.pdf	26/08/2025 17:57	WPS PDF 文档
3.6 Action Group Editing – Feature Overvi...	26/08/2025 16:53	DOCX 文档
3.6 Action Group Editing – Feature Overvi...	26/08/2025 17:57	WPS PDF 文档
3.7 Execute and Download Actions.docx	26/08/2025 17:00	DOCX 文档
3.7 Execute and Download Actions.pdf	26/08/2025 17:57	WPS PDF 文档
3.8 Action Editing.docx	26/08/2025 17:05	DOCX 文档
3.8 Action Editing.pdf	26/08/2025 17:57	WPS PDF 文档
3.9 Integrating Action Files.docx	26/08/2025 17:16	DOCX 文档
3.9 Integrating Action Files.pdf	26/08/2025 17:58	WPS PDF 文档
3.10 App Custom Control.docx	18/08/2025 14:47	DOCX 文档
3.10 App Custom Control.pdf	26/08/2025 18:20	WPS PDF 文档

4. **Arduino IDE Installation & Servo Calibration:** Arduino IDE: Includes installation package, user manual, testing demos, and interface introduction. Servo Calibration: Provides methods and standards for servo deviation calibration and tuning.

Name	Date modified	Type
4.1 Introduction and Installation of the Ar...	20/08/2025 15:21	File folder
4.2 Servo Calibration	28/08/2025 18:20	File folder

5. **Motion Control Basic Course:** Includes sample code and documentation for fundamental robot movement. Covers inverse kinematics concepts and gait overview. By learning this section, you can programmatically control the robot's movement and posture.

Name	Date modified	Type
5.2 Omnidirectional Movement	20/08/2025 15:20	File folder
5.3 Turn Left and Right	20/08/2025 15:20	File folder
5.4 Speed Adjustment	20/08/2025 15:20	File folder
5.5 Gait Parameter Adjustment	20/08/2025 15:20	File folder
5.6 Posture Adjustment	20/08/2025 15:20	File folder
5.1 Kinematics and Gait Overview.docx	15/08/2025 11:25	DOCX 文档

6. **Secondary Development Course:** Provides example projects using built-in or external sensors, source code for motion group calls, and in-depth code analysis. Covers sensor-based interactive features and continuous motion group execution.

Name	Date modified	Type
6.1 Action Group Overview and Hands-o...	20/08/2025 15:36	File folder
6.2 Intelligent Voice Control	20/08/2025 15:36	File folder
6.3 Ultrasonic Ranging	20/08/2025 15:36	File folder
6.4 Ultrasonic Obstacle Avoidance	20/08/2025 15:36	File folder
6.5 Automatic Following	20/08/2025 15:36	File folder
6.6 Self-Balancing	20/08/2025 15:36	File folder
6.7 Dot Matrix Display	20/08/2025 15:36	File folder
6.8 Ultrasonic Ranging and Displaying	23/08/2025 14:30	File folder
6.9 Touch Control	20/08/2025 15:36	File folder
6.10 Infrared Obstacle Avoidance	20/08/2025 15:36	File folder
6.11 Intelligent Fall Prevention	20/08/2025 15:36	File folder

7. **AI Vision Feature Course:** Includes sample source code and detailed documentation for the ESP32-S3 vision module. Teaches how to implement color recognition, color tracking, face detection, and more through programming.

Name	Date modified	Type
7.1 ESP32-S3 AI Vision Module	20/08/2025 16:02	File folder
7.2 Color Recognition	23/08/2025 14:32	File folder
7.3 Color Tracking	20/08/2025 16:33	File folder
7.4 Vision Line Following	20/08/2025 16:02	File folder
7.5 Face Recognition	20/08/2025 16:02	File folder

8. **AI Voice Interaction Course:** Includes sample source code and

documentation for the WonderEcho voice module. Teaches how to implement human–robot interaction, voice control, and intelligent speech recognition.

Name	Date modified	Type
8.1 Introduction and Installation of Wond...	20/08/2025 16:02	File folder
8.3 Voice Obstacle Alert	20/08/2025 16:02	File folder
8.4 Human-Robot Interaction	20/08/2025 16:02	File folder
8.5 Voice Control	20/08/2025 16:02	File folder
8.2 Voice Module Library Files Introductio...	19/08/2025 10:39	DOCX 文档

9. **Serial Communication Hands-on Course:** Introduces the principles of master-slave communication and the robot's serial communication protocol. By completing this course, you can control the robot through serial communication.

Name	Date modified	Type
9.2 PC Serial Control	20/08/2025 16:02	File folder
9.1 Communication Principles Between ...	19/08/2025 14:16	DOCX 文档
9.3 miniHexa Host & APP Communicati...	19/08/2025 15:03	WPS PDF 文档

3. Learning Path

To help you learn and master robotics efficiently, we recommend following the progressive learning path below:

Stage 1: Getting to Know the Robot (Recommended First)

Chapter 1: Read First (Key Content)

- Learn the robot's basic functions, interface descriptions, usage tutorials, and important precautions.

Stage 2: Controlling the Robot (Mastering Basic Control)

1. Chapter 2: App Control

- Learn about the robot's motion types, postures, features, and control methods.

2. Chapter 3: PC Control and Action Group Editing

- Learn how to use the PC software and edit the robot's action groups.

Stage 3: Setting Up the Development Environment & Debugging

1. Chapter 4: Arduino IDE Installation & Servo Calibration (Key Content)

- Learn how to set up the development environment, use the IDE interface, and perform robot deviation debugging.

Stage 4: Program Control

1. Chapter 5: Motion Control Basic Course

- Learn how to control the robot's movements and postures through core control programs.

2. Chapter 6: Secondary Development Course

- Learn how to expand features using programming and built-in or external sensors, call robot action groups, and analyze source code in detail. Covers sensor-based interactive features and continuous motion group execution.

3. Chapter 7 AI Vision Feature Course & Chapter 8 AI Voice Interaction Course

- Learn how to use programming with vision and voice modules to enable visual and voice-based interaction.

4. Chapter 9: Serial Communication Hands-on Course

- Learn how to control the robot via serial communication using program code.

Stage 5: Extended Learning

- After completing the core robotics courses, you may deepen your knowledge further. If you want to understand hardware and circuit principles, study the materials in folder **3. Technical Parameter & Diagram**.